

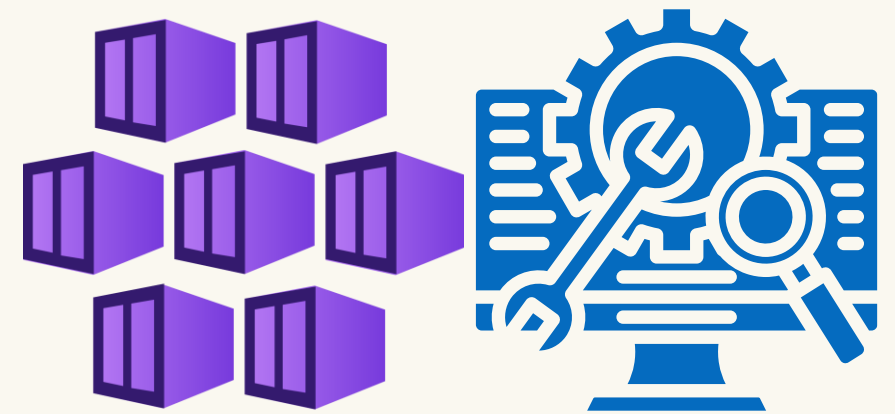
Resource reservations

and

Kube-reserved
resource optimization

in

Azure Kubernetes
Service (AKS)



Today, we'll learn and cover:



1.

Resource
reservation
considerations

2.

Reserved CPU

3.

Reserved
memory
(K8s versions 1.28
and lower)

4.

Kube-reserved
optimization:
Reserved memory
(K8s versions 1.29
and higher)

5.

Confusions the
resource
reservations can
lead to

1. Resource reservation considerations

The goal

AKS reserves resources from the nodes to ensure proper functionality and to avoid competing for resources between pods, system daemons (like kubelet and containerd), and node's operating system



!

Resource reservations lead to a **discrepancy** between the total and allocatable resources

!

It is important to be aware of resource reservations, especially when **requests and limits** are set

!

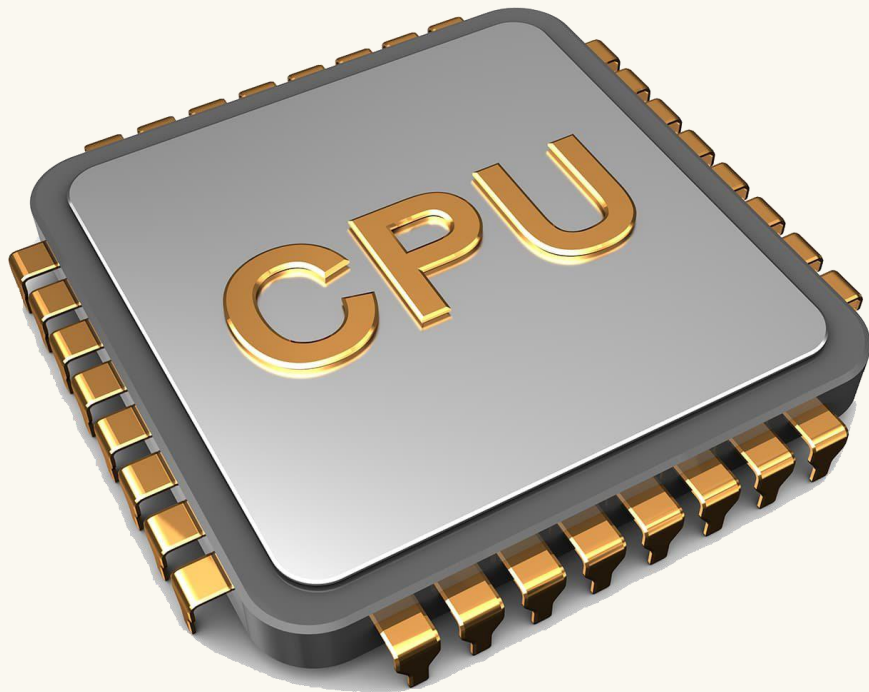
The resource reservations **depend on multiple factors**, especially node size

!

The resource reservations is set with the help of **"kubeReserved"** kubelet parameter

2.

Reserved CPU



CPU Cores	Kube-reserved (milicores)
1	60
2	100
4	140
8	180
16	260
32	420
64	740

3. Reserved memory (K8s versions 1.28 and lower)

The memory reservation is the sum of:

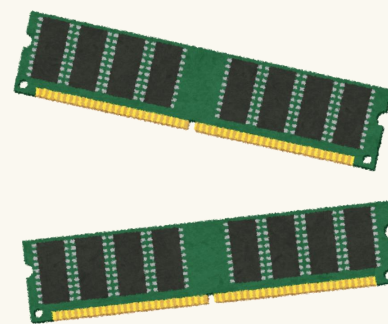
- kubelet daemon
- regressive rate reservations

The **kubelet daemon** manages the container creation and termination and requires at least **750 Mi** of allocatable memory



The **regressive rate** of memory reservation is for the **Kubelet daemon** to run effectively

2 more GB are reserved for system process in Windows nodes



Percentage	Amount of
25%	the first 4 GB of memory
20%	the next 4 GB of memory (up to 8 GB)
10%	the next 8 GB of memory (up to 16 GB)
6%	the next 112 GB of memory (up to 128 GB)
2%	any memory above 128 GB

4. Kube-reserved optimization: Reserved memory (K8s versions 1.29 and higher)

The memory reservation is the sum of:

- kubelet daemon
- rate of memory reservation

The **kubelet daemon** requires at least **100 Mi** of allocatable memory



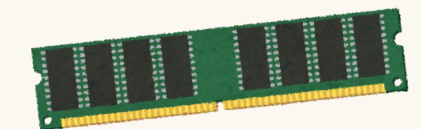
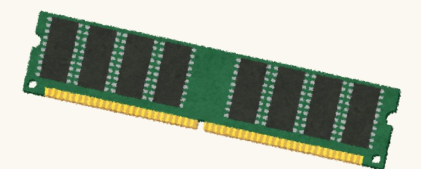
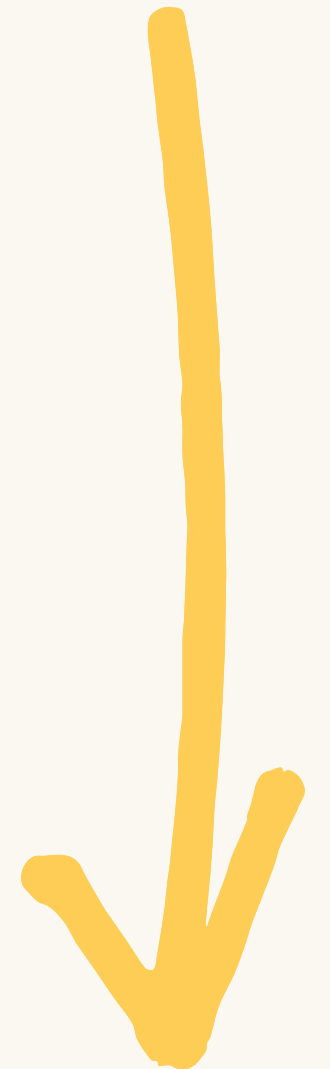
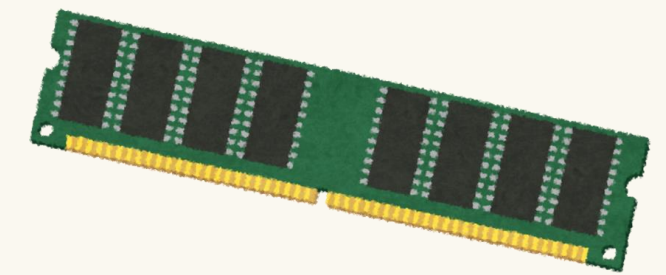
The **rate of memory reservation** is the lesser value of:

- $20\text{MB} * \text{Max Pods supported on the Node} + 50\text{MB}$

OR

- 25% of the total system memory resources

2 more GB are reserved for system
process in Windows nodes



5. Confusions the resource reservations can lead to

!

“My AKS nodes cannot use their real capacity”

!

“I have a pod with x requests, the node has more CPU/memory available, but the pod is stuck in Pending”

!

“Cluster autoscaler/HPA doesn't scale”

